

■ 2.6.14 Special Topic: External Programs

Mathematica does many things well. However, there are some things that of convenience or necessity are best done by external programs.

On most computer systems (e.g. those running UNIX), you can run external programs from within *Mathematica*.

Mathematica provides a number of methods for “communicating” with external programs. All the methods that are currently supported ultimately involve the exchange of plain text between *Mathematica* and the external program.

Use files which can be read and written both by *Mathematica* and by an external program.

Use `<<"!command"` and `>>"!command"` to read or write expressions directly to an external program.

Use `RunThrough` to pass expressions through an external program.

Use `Read` with `"!command"` to read objects produced by an external program.

Use `StartProcess` and `CallProcess` to access individual functions in a suitably-prepared external program.

Ways to communicate with external programs in *Mathematica*.

Mathematica allows you to treat communication with external programs in much the same way as you treat input and output to files. Thus a command like `expr>>string` can be used either to write output to a file, or to send output to an external program.

If *string* is a standard file name, then `expr>>string` simply writes the text of *expr* to the file. If, on the other hand, *string* starts with a `!`, then it is interpreted as a command for running an external program. In this case, the textual form of *expr* is fed through a “pipe” as the input to the external program. Thus, for example, `expr>>lpr` writes *expr* to a file named *lpr*. `expr>>!lpr`, on the other hand, executes the external program *lpr*, giving the textual form of *expr* as its input. (On a typical UNIX system, *lpr* sends its input to a printer; as a result, `expr>>!lpr` causes a *Mathematica* expression to be printed.)

<code>expr>>filename</code>	write an expression to a file
<code>expr>>"!command"</code>	send an expression through a pipe as input to an external program

Writing expressions to files and external programs.

On most computer systems, the text that you give after `!` is used directly as an operating system command. Since this text often contains spaces or other special characters, you will usually have to enclose `!command` in `"` when you give it in *Mathematica*.

You can use `!command` wherever you would usually use a standard file name. As well as sending *Mathematica* output to external programs using `>>!command`, you can also get *Mathematica* input from an external program using `<<!command`.

`echo $TERM` is a UNIX operating system command that prints the type of terminal you are using. The text output from this command is read as input to *Mathematica*.

```
In[4]:= <<"!echo $TERM"
Out[4]= vt100
```